संगणक विज्ञान एवं अभियांत्रिकी विभाग
Department of Computer Science & Engineering
भारतीय प्रौद्योगिकी संस्थान जोधपुर
Indian Institute of Technology Jodhpur
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Indian Institute of Technology Jodhpur

M.Tech/ M.Des./ M.Sc./Master's Program Course Curriculum Department of Computer Science and Engineering

Course Structure: Semester wise distribution of credits for M.Tech (CSE) students for the batches from Academic Year 2025-26.

Cat.	Course Number, Course Title		L-T-P	Credits		Cat.	Course Number, Course Title		L-T-P	Credits
I Semester						**II Semester**				
C	CSL7130	Mathematical Foundations for Computer Science	3-0-0	3		C	CSL7090	Software & Data Engineering	3-0-0	3
C	CSL7560	Advanced Data Structures and Algorithms	3-0-0	3		E	XXX7XX0	Elective	3-0-0	3
C	CSL7190	Advanced Architecture and OS	3-0-0	3		E	XXX7XX0	Elective	3-0-0	3
E	XXX7XX0	Elective	3-0-0	3		E	XXX7XX0	Elective	3-0-0	3
NG1	HSN7XX0	Non-Graded I	1-0-0	S/X		NG2	HSN7XX0	Non-Graded II	1-0-0	S/X
Total				**12**		Total				**12**
T	CSS7000	Summer Thesis								8
III Semester						**IV Semester**				
T	CST7010	Thesis		16		T	CST7XXX	Thesis		16
Total				**16**		Total				**16**

Reduced Electives CS

S.No.	Course Number	Course Title		
1	CSL7150	Cybersecurity	3-0-0	3
2	CSL7140	Complexity Theory	3-0-0	3
3	CSL7160	Advanced Algorithms	3-0-0	3
4	CSL7XXX	Emerging Network Technologies	3-0-0	3
5	CSL7XXX	Blockchain Technologies	3-0-0	3
6	CSL7360	Computer Vision	3-0-0	3
7	CSL7750	Distributed Database Systems	3-0-0	3
8	CSL7510	Virtualisation and Cloud Computing	3-0-0	3
9	CSL7620	Machine Learning	3-0-0	3
10	CSL7460	Mobile and Pervasive Computing	3-0-0	3
11	CSL7640	Natural Language Understanding	3-0-0	3
12	CSL7XX0	High Performance Computing Architectures	3-0-0	3

S.No.	*Category*	*Course Category Title*	*Total Courses*	*Total Credits*
1	C	*COMPULSORY*	4	12
2	E	*ELECTIVES*	4	12
3	NG	*Non-Graded*	2	0
3	T	Thesis	**2 (16+16)**	32
		Total		**56**

Department of Computer Science and Engineering

M.Tech (CSE) Courses

1. The following four courses are compulsory for M.Tech. students of the department:

   1. **Mathematical Foundations for Computer Science**
   2. **Advanced Data Structures and Algorithms**
   3. **Advanced Architecture and OS**
   4. **Software & Data Engineering**

2. M.Tech. students shall have the option to choose four courses from the following list of courses:

   1. Cybersecurity
   2. Complexity Theory
   3. Advanced Algorithms
   4. Emerging Network Technologies
   5. Blockchain Technologies
   6. Computer Vision
   7. Distributed Database Systems
   8. Virtualisation and Cloud Computing
   9. Machine Learning
   10. Mobile and Pervasive Computing
   11. Natural Language Understanding
   12. High Performance Computing Architectures

The syllabus and checklists for the new proposed courses are given below:

Title	**Advanced Data Structures & Algorithms**	Number	**CS7560**
Department	Computer Science	L-T-P [C]	3–0–0 [3]
Offered for	M.Tech. CSE and M.Tech. AI	Type	core
Prerequisite	Algorithm Design and Analysis, Data Structures, Maths for Computing		

Objectives

1. The objective of the course is to introduce several advanced algorithmic techniques and data structures.
2. Implementation of various advanced data structure and algorithms to solve real life problems

Learning Outcomes

1. Learn advanced data structure techniques
2. Learn a new set of techniques to cope with NP-hard problems.
3. Identify novel and significant open research questions in the field.
4. Implementation of various advanced data structure and algorithms to solve real life problems
5. Understand the intractability of problems.

Contents

Data Structures and Algorithms Techniques (Fractals I and II)

Advanced Data Structures [10 lectures]: Red-black trees, Interval trees, Dynamic order statistics, Binomial heap, Fibonacci Heap [5 lectures], Tries, Splay Trees, B-Trees, Hashing - disjoint set, union find [5 lectures].

Network Flows [5 lectures]: Maximum Flow problem, Ford-Fulkerson algorithm, Max flow min cut theorem, augmenting paths, applications to bipartite matching problem and disjoint paths in directed and undirected graphs.

Amortized Analysis [3 lectures]: Aggregate Analysis, Accounting method, Potential method.

Dynamic Programming [6 lectures]: Knapsack, Independent Set of Trees, Interval Scheduling, Set Cover, Permutation Problems, Partition Problems, String matching: naive method, Knuth-Morris-Pratt algorithm.

NP-completeness and Reductions [8 lectures]: Complexity classes P, NP, co-NP, NP-complete, NP-Hard, Example reductions between problems.

Fractal III: Advanced Algorithmic Techniques

Advanced Algorithmic Techniques [10 lectures]: Approximation Algorithms: Greedy Algorithm – Load Balancing, Set Cover [2 Lectures]; The Pricing Method: Vertex Cover, Knapsack [3 Lectures], Randomized Algorithms [5 Lectures].

Textbooks

1. Jon Kleinberg, Eva Tardos (2005), Algorithm Design, Pearson Education, 1st Edition.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms.

Self Learning Material

1. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-046j-introduction-to-algorithms-sma-5503-fall-2005/video-lectures/
2. https://nptel.ac.in/courses/106/101/106101060/
3. https://nptel.ac.in/courses/106/104/106104019/

Title	**Advanced Architecture and Operating Systems**	Number	CSL7190
Department	CSE	L-T-P[C]	3–0–0 [3]
Offered for	M.Tech. CSE and M.Tech. AI	Type	Elective
Prerequisite	UG level OS and Computer Organization or Architecture	Antirequisite	None

Objectives

1. To develop foundations in advanced aspects of modern operating systems and architectures.
2. To develop capabilities in design thinking and systems design centered around the core layers of the computer.
3. To develop knowledge on end layer application use cases, performance, and usability.

Contents

Architecture: [24 lectures]

Processor Design [11 lectures]: Instruction Set Architecture (ISA) design, Advanced Pipelining, ILP, Out of order pipelines, Branch Prediction, Static and Dynamic Scheduling, Hardware Multithreading, Multicore Processors, Accelerators (ASIC/FPGA/GPU platforms).

Memory System: Memory Hierarchy [11 lectures], Advanced Cache Design, Instruction and data prefetching, Cache Coherence Protocols, Memory Consistency, Synchronization, non-uniform caches, Main memory design, Emerging Memory Technologies.

On-Chip Networks [2 lectures]: Message transmission, routing, performance.

Operating System: [15 lectures]

OS Fundamentals review, Types of OS, Types of Kernels, Design Approach, Virtualization (hardware-assisted), Containers and VM, Virtual machines scheduling,

Advanced Memory Management, Handling large pages, TLB-Memory Hierarchy Interactions, NUMA, Advanced File Systems, Distributed Operating Systems

Concurrency and Transactions, Security.

Textbooks

1. Arpaci-Dusseau, Remzi H., C. Andrea, and William Stallings. "Operating Systems: Three Easy Pieces."
2. Mukesh Singhal and Niranjan G. Shivaratri, “Advanced Concepts in Operating Systems Distributed, Database, and Multiprocessor Operating Systems”, Tata McGraw-Hill.
3. Hennessy, John L., and David A. Patterson. Computer architecture: a quantitative approach. Elsevier.

Reference

1. Research papers and reading material provided by instructors.

Title	**Software and Data Engineering**	Number	CSL7090
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M.Tech., PhD	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Discuss techniques to manage a large amount of data
2. Provide mechanisms to design and develop data-intensive computing systems

Learning Outcomes

The students are expected to have the ability to:

1. Design complex end-to-end data pipeline for data processing
2. Critically identify and use the tools for data handling and management
3. Use modern software technologies to design and develop data analytical systems

Contents

CSL7XX0 Cloud Computing and Virtualization 1-0-0 [1]

Basics of complex software design: Concept of modular software, microservices, communication, 4+1 architectural views and patterns (5 lectures)

Cloud Computing: Architecture of cluster computing, design of data centers, open data center platforms, fault-tolerant system design (5 lectures)

Virtualization: Type-1 and Type-2 virtualization, virtual machine, containers, dockers (4 lectures)

CSL7XX0 Data Management 1-0-0 [1]

Data Management: Structured data, relational database management, unstructured data, semi-structured data, Nosql database management (mongodb), column database, graph database, XML, JSON, HDFS, Handling drift in data, sensor data reliability at software and algorithmic level, sensor data analysis techniques (14 lectures)

CSL7XX0 Data Intensive Processing Systems 1-0-0 [1]

Data Intensive Processing Systems: Architecture of large scale data processing systems, Hadoop, Apache Spark, Storm, parallel data processing concepts such as map-reduce, directed acyclic graph, resilient distributed datasets, dynamic resource allocation, partial & shared computation, storage architecture (14 lectures)

Textbook

1. Bass L., Clements P., Kazman R., (2012), *Software Architecture in Practice*, 3rd edition, Addison-Wesley Professional
2. Martin K., (2017), *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st Edition, O'Reilly Media

Self Learning Material

1. Tylor,R.N., Medvidovic,N. and Dashofy,E.M., (2014), *Software Architecture Foundation: Theory and Practice*, Wiley

Preparatory Course Material

1. IEEE Transactions on Knowledge and Data Engineering
2. International Conference on Data Engineering

Course Title	**Cybersecurity**	Course No.	**CSL7150**
Department	Computer Science and Engineering	Structure (L-T-P [C])	3-0-2 [4]
Offered for	B. Tech, MTech, Ph. D.	Type	Elective
Prerequisite	Computer Networks	Anti-req	Security and its applications

Objectives

1. The Instructor will provide the skills needed to protect networks, secure electronic assets, prevent attacks, ensure the privacy of your customers, and build secure infrastructure

Learning Outcomes

1. To protect data and respond to threats that occur over the Internet
2. Design and implement risk analysis, security policies, and damage assessment
3. An ability to apply security principles and practices to the environment, hardware, software, and human aspects of a system.

Contents:

Introduction to Cyber Security: *Internet governance – Challenges and constraints, Cyber threats. (2 Lectures)*

Cyber Security Vulnerabilities and Cyber Security Safeguards: *Cyber security, Vulnerabilities, safeguards, Access control, Authentication, Biometrics, Deception, Denial of Service Filters, Ethical hacking, Firewalls, Response, Scanning, Security policy, Threat management. (8 Lectures)*

Cryptography: *Shannon's Approach to Cryptography: Measures of security, Perfect secrecy, Definition of entropy, One-time pad , Symmetric Key Cryptography, Cryptographic Hash Functions, Authentication, Public Key Cryptosystems, Key Distribution and Key Agreement Protocols(4 Lectures)*

Securing Web Application, Services and Servers: *Basic security for HTTP applications and services, Basic security for SOAP services, Identity management and Web services, Authorization patterns, Security considerations, challenges. (4 Lectures)*

Network Security: *TCP/IP threats, The IPSEC protocol, The SSL and TLS protocols, Firewalls and Virtual Private Networks (VPNs), Electronic mail security, Worms, DDoS attacks, BGB and security considerations (12 lectures)*

Intrusion Detection and Prevention: *Intrusion detection and Prevention techniques, Anti-malware software, Security information management, Network session analysis, System integrity validation. (4 Lectures)*

Cyberspace and the Law: *Cyber security regulations, Roles of international law, State and Private Sector in cyberspace, Cyber security standards. The INDIAN cyberspace, National cyber security policy. (5 Lectures)*

Cyber Forensics: *Handling preliminary investigations, Controlling an investigation, Conducting disk-based analysis, Investigating information-hiding, Tracing internet access, Tracing memory in real-time. (4 Lectures)*

Laboratory

1. Design and implementation of a simple client/server model and running application using sockets and TCP/IP.
2. To make students aware of the insecurity of default passwords, printed passwords and passwords transmitted in plain text.
3. To teach students how to use SSH for secure file transfer or for accessing local computers using port forwarding technique.
4. Comparison between Telnet and SSH for Secure Connection
5. AVISPA Tool for the Automated Validation of Internet Security Protocols and Applications

Text Book

1. C.J. HOOFNAGLE (2016), Federal Trade Commission Privacy Law and Policy, Cambridge University Press, 2016.
2. Stallings, W. (2017). Cryptography and network security, 7/E. Pearson Education India

Reference Book

1. Douglas R. Stinson, Maura B. Paterson (2018). Cryptography: theory and practice. 4/E Chapman and Hall/CRC
2. P.W. SINGER, A. FRIEDMAN (2014), Cybersecurity: What Everyone Needs to Know, OUP, 1st Edition.

Supplementary Resource

Title	**Complexity Theory**	Course No.	**CSL7140**
Department	Computer Science and Engineering	Structure (L-T-P [C])	3–0–0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CSE, AI), M.Sc. (Maths)	Type	Core
Prerequisite	Design and Analysis of Algorithms, Maths for Computing (or Discrete Mathematics)	Antirequisite	

Objectives:

1. Learning the limits of computation.
2. Learning basic complexity classes and various computational models.

Learning Outcomes:

1. A Complexity Theory course covers computational complexity and how to analyze the time and space efficiency of algorithms using big-O notation. Students learn about complexity classes like P, NP, NP-hard, and NP-complete, and how to establish problem hardness using reduction techniques.
2. The course introduces complexity hierarchies, alternation, probabilistic complexity. Students gain skills in classifying problems and understanding the limits of efficient computation.

Contents

Computability [7 lectures]: Review of Turing Machine (TM) and its variants, Universal Turing Machines and encoding TMs, Uncomputability, Complexity Class P.

Time Complexity [7 lectures] : Time as a resource, the Complexity Class NP, P vs NP, NP-completeness, Cook-Levin Theorem, Reductions.

Diagonalization [3 lectures]: Time-hierarchy theorem, Ladner's theorem, Mahaney's theorem.

Space Complexity [7 lectures]: Space as a resource, the Complexity Classes PSPACE, L and NL, Reachability Problem, PSPACE and NL completeness, Space Hierarchy Theorem, Savitch's theorem, Immerman-Szelepscenyi theorem.

Polynomial Hierarchy [4 lectures]: The Polynomial Hierarchy, Alternating TMs, Time-Space Tradeoffs for the Satisfiability problem.

Circuits [7 lectures]: Boolean Circuits and the Class P/poly**,** Advice taking TMs, Karp-Lipton Theorem, Circuit Lower Bounds, the Complexity Classes NC and AC.

Randomized Computation [7]: Probabilistic TMs, the Complexity Classes RP, coRP and ZPP, Error-reduction in randomized computation, Sipser-Gacs theorem, Randomized bounded-space computation.

Textbooks

1. S. Arora and B. Barak (2009), Complexity Theory: A Modern Approach, Cambridge University Press.

Reference books

1. Sipser, M., (2013), Introduction to the Theory of Computation, Cengage Learning.
2. Oded Goldreich (2008), Computational Complexity: A Conceptual Perspective.

Self Learning Material

1. https://www.youtube.com/playlist?list=PLm3J0oaFux3YL5vLXpzOyJiLtqLp6dCW2
2. https://onlinecourses.nptel.ac.in/noc21_cs49/preview

Title	**Advanced Algorithms**	Number	**CSL7160**
Department	Computer Science	L-T-P [C]	3–1–0 [4]
Offered for	B.Tech., M.Tech., Ph.D. (CSE, AI), M.Sc. (Maths)	Type	Elective / Core Specialization
Prerequisite	Design and Analysis of Algorithms, Maths for Computing (or Discrete Mathematics)		

Objectives

1. The objective of the course is to introduce several advanced algorithmic techniques such as parameterized algorithms, approximation algorithms, randomized algorithms, etc.

Learning Outcomes

1. Learn a new set of techniques to cope with NP-hard problems.
2. Identify novel and significant open research questions in the field.

Contents

Parameterized Algorithms [6 lectures]: Introduction to Parameterized Complexity and basics techniques: Branching, Iterative Compression.
Approximation Algorithms: [10 lectures]: Greedy Algorithm – Load Balancing, Center Selection Problem, Set
Cover [5 Lectures]; The Pricing Method: Vertex Cover, Linear Programming and Rounding: An application to Vertex Cover, Knapsack [5 Lectures]
Randomized Algorithms [10 lectures]: Contention Resolution, Global, Random Variables and Expectations, Max-3-SAT approximation [7 Lectures]; Color Coding [3 Lectures]
Exact Exponential Time Algorithms [7 lectures]: Exact Algorithms for Coloring, SAT, Directed Feedback Arc Set, Max-Cut, Monotone-Local-Search, Or some other topics of contemporary interest.
Streaming Algorithms [5 lectures]: Introduction to streaming algorithms and its application to some graph theoretic problems.
Online Algorithms [4 lectures]: Introduction to online algorithms: definition, the difference with offline setting; list update problem, paging problem, Bin packing.

Textbooks

1. Marek Cygan, Fedor V. Fomin, Lukasz Kowalik, Daniel Lokshtanov, Daniel Marx, Marcin Pilipczuk, Michal Pilipczuk, Saket Saurabh (2015): Parameterized Algorithms, Springer.
2. Jon Kleinberg, Eva Tardos (2005), Algorithm Design, Pearson Education, 1st Edition.
3. Fedor V. Fomin, Dieter Kratsch (2010), Exact Exponential Time Algorithms, An EATCS Series, Springer.

Self Learning Material

1. https://www.youtube.com/watch?v=Ex8TueBsF1g&list=PLhkiT_RYTEU0gpi97fqjtaHy9Gk47oF85&index=1
2. https://sites.google.com/view/sakethome/teaching/parameterized-complexity?authuser=0
3. https://www.youtube.com/watch?v=S8Acu3EpvsE&list=PLhkiT_RYTEU2itsMqCNdXUg4cdFUWJn3-&index=4
4. https://www.youtube.com/watch?v=jNfQ3GZlrjM&list=PLhkiT_RYTEU3vSaVleEm_-bIPBzCqRQHK
5. https://www.youtube.com/watch?v=0rB6Pyq2cLQ&list=PLTtM9ThZ2L-c638AjqTskivmXwVTzTYnl&index=1

Title	**Blockchain Technology**	Course No.	CSL 7XX
Department	Computer Science and Engineering	Structure (L-T-P [C])	3–0–0 [3]
Offered for	B.Tech, MTech, PhD.	Type	Elective
Prerequisite	Cryptography	Antirequisite	

Objectives

The Instructor will:

1. Explain how blockchain technology works
2. Integrate blockchain technology into the current business processes to make them secure

Learning Outcomes

1. Understand what and why of Blockchain
2. Explore major components of Blockchain and Identify a use case for a Blockchain application Create their own Blockchain network application

Contents

Introduction to Blockchain: Digital Trust, Asset, Transactions, Distributed Ledger Technology, Types of network, Components of blockchain, mining, double spending attack, 51% attack. (5 Lectures)
PKI and Cryptography: Private keys cryptography, Public keys cryptography, Hashing, Digital Signature. (6 Lectures)
Consensus: Byzantine Fault, Proof of Work, Proof of Stake. (5 Lectures)
Cryptocurrency: Bitcoin creation and economy, Limited Supply and Deflation, Hacks, Ethereum concept and Ethereum classic, Monero, Zcash, Zero knowledge proofs, etc. (9 Lectures)
Permissioned blockchains: Permissionless versus permissioned Blockchains (2 Lectures)
Hyperledger Fabric: Hyperledger Architecture, Membership, Blockchain, Transaction, Chaincode, Hyperledger Fabric, Features of Hyperledger, Fabric Demo. (8 Lectures)
Blockchain Applications: Building on the Blockchain, Ethereum Interaction - Smart Contract and Token (Fungible, non-fungible), Languages,Payment escrow, Micro payments, Decentralized lotteries Blockchain-as-a-service. (6 Lectures)

Textbook

A. BAHGA, V. MADISETTI (2017), Blockchain Applications: A Hands-On Approach, VPT.

Self Learning Material

1. M. SWAN (2015), Blockchain: Blueprint for a New Economy, O'Reilly Media.
2. R. WATTENHOFER (2016), The Science of the Blockchain, CreateSpace Independent Publishing Platform.
3. I. BASHIR (2017), Mastering blockchain, Packt Publishing Ltd.
4. K.E. LEVY, Book-smart, Not Street-smart: Blockchain-based Smart Contracts and the Social Workings of Law, Engaging Science, Technology, and Society, vol. 3, pp. 1-15, 2017.

Preparatory Course Material

MIT Online Blockchain Course, Learn Blockchain Technology: https://getsmarter.mit.edu/

Title	**Computer Vision**	Number	CSL7360
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech, M.Tech., PhD	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. Provide insights into fundamental concepts and algorithms behind some of the remarkable success of Computer Vision
2. Impart working expertise by means of programming assignments and a project

Learning Outcomes

The students are expected to have the ability to:

1. Learn and appreciate the usage and implications of various Computer Vision techniques in real-world scenarios
2. Design and implement basic applications of Computer Vision

Contents

Introduction: The Three R's - Recognition, Reconstruction, Reorganization (1 lecture)
Perspective: Static Perspective, Transformations, Dynamic perspective (5 lectures)
Fundamentals of Image formation and processing: Radiometry of image formation, Basic image processing, Biological visual processing (5 lectures)
Recognition: Object recognition case study - identifying digits with multiple approaches, Visual grouping, Convolutional Neural Network (ConvNet) based approaches to visual recognition of objects and scenes, Deformable Parts Model (DPM), Attributes, pose and actions (8 lectures)
Analysis: Binocular Stereopsis, Markov Random Fields in Computer Vision, Solving for stereo correspondence, Optical flow, Review of differential geometry (12 lectures)
Detection and Segmentation: Contour detection, Bottom-up segmentation, Gestalt grouping heuristics, Semantic segmentation - instance segmentation and pixel classification, Pose and keypoint estimation (5 lectures)
Image understanding: Scene understanding from RGBD images, 3D perception from a single image, Face recognition (6 lectures)

Textbook

1. Hartley,R. Zisserman,A., (2004), *Multiple View Geometry in Computer Vision*, 2nd Edition, Cambridge University Press
2. Szeliski,R., (2010), *Computer Vision: Algorithms and Applications*, Springer-Verlag London

Reference Books

1. Research literature

Course Title	**Distributed Database Systems**	Course No.	**CSL7750**
Department	Computer Science & Engineering	Structure (L-T-P [C])	3-0-0 [3]
Offered for	B.Tech, MTech, PhD	Type	Elective
Prerequisites	Operating Systems, Database Systems, Computer Networks	Antirequisite	None

Objective
To understand and appreciate concepts of distributed database design, and its associated issues of consistency, concurrency, optimization, integrity, reliability, privacy, and security.

Learning Outcome
Ability to understand the need for distributed database systems and its related complexities pertaining to fragmentation, replication, availability, concurrency, consistency and recovery.

Contents
Introduction: Distributed data processing concepts, What is a DDBS - advantages, disadvantages and problem areas. (2 Lectures)
Distributed Database Management System Architectures: Transparencies, Architecture, Global directory concepts and issues. (3 Lectures)
Distributed Database Design: Design strategies, Design issues, Fragmentation, Data allocation. (4 Lectures)
Semantics Data Control: View management, Data security, Semantic integrity control. (5 Lectures)
Query Processing: Objectives, Characterization of processors, Layers of processing, Query decomposition, Data localization. (5 Lectures)
Query Optimization: Factors, Centralized query optimization, Fragmented query ordering, Query optimization algorithms. (5 Lectures)
Transaction Management: Goals, Properties, Models. (4 Lectures)
Concurrency Control: Concurrency control in centralized systems, Concurrency control in DDBSs - algorithms, Deadlock management. (5 Lectures)
Reliability: Issues and types of failures, Reliability techniques, Commit protocols, Recovery protocols. (5 Lectures)
Other Avenues: Parallel database Systems, Multi-databases. (4 Lectures)

Reference Books
S. CERI, G. PELAGATTI (2008), Distributed Databases: Principles and Systems, McGraw-Hill, 1st Edition (2017 Reprint).
M.T. ÖZSU, P. VALDURIEZ (2011), Principles of Distributed Database Systems, Springer, 3rd Edition.

Course Title	**Virtualization and Cloud Computing**	Course No.	**CSL7510**
Department	Computer Science & Engineering	Structure (L-T-P [C])	3-0-0 [3]
Offered for	B.Tech	Type	Elective
Prerequisite	Operating Systems, Computer Networks	Antirequisite	None

Objectives
This course will introduce fundamentals of virtualization and concepts, frameworks, and applications in Cloud Computing to the audience

Learning Outcomes
1. To understand various virtualization techniques and their features and limitations
2. To understand design and concepts of a cloud computing framework

Contents
Cloud Computing: Concept, Definition, Cloud Types and Service Deployment Models. (5 Lectures)
Virtualization: Concept, Definition, Types of Virtualization, Hardware Virtualization, Full and Para Virtualization, Hypervisors, Hardware-assisted virtualization, operating system level virtualization, application virtualization (10 Lectures)
Virtual and Physical Networking: Introduction, vSwitches, virtual NICs, Virtual Networking, virtual LAN (5 Lectures)
Storage Virtualization: Introduction, SAN/NAS versus storage virtualization. (4 Lectures)
Virtual Machine Management: Base Virtual Machine, Virtual CPUs, Sockets, Cores, Memory Scaling Up and Scaling Down, USB Support, Virtual Disks, Live Migration. Security. (10 Lectures)
Containers: Concept, Definition, Docker, Container versus Virtualization, Portability, Remote deployment (5 Lectures)
Applications and Case Studies: Linux KVM, VirtualBox, Openstack. (3 Lectures)

Text Book
1. D.E. SARNA (2010), Implementing and Developing Cloud Computing Applications, CRC Press.
2. B.S. SODHI (2017), Topics in Virtualization and Cloud Computing, Ropar PB India, 2017.

Reference Book
B. FURHT, A. ESCALANTE (2010), Handbook of Cloud Computing (Vol. 3), Springer.

Title	**Machine Learning**	Number	CSL7620
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M.Tech. (CSE, AI, DCS), PhD	Type	Compulsory
Prerequisite	Introduction to Computer Sc., Probability, Statistics and Stochastic Processes, Linear Algebra	Antirequisite	IML, PRML

Objectives

1. To understand various key paradigms for machine learning approaches
2. To familiarize with the mathematical and statistical techniques used in machine learning.
3. To understand and differentiate among various machine learning techniques.

Learning Outcomes

The students are expected to have the ability to:

1. To formulate a machine learning problem
2. Select an appropriate pattern analysis tool for analyzing data in a given feature space.
3. Apply pattern recognition and machine learning techniques such as classification and feature selection to practical applications and detect patterns in the data.

Contents

Fractal I: Supervised Learning

Introduction: Definitions, Datasets for Machine Learning, Different Paradigms of Machine Learning, Data Normalization, Hypothesis Evaluation, VC-Dimensions and Distribution, Bias-Variance Tradeoff, Regression (Linear) (7 Lectures)

Bayes Decision Theory: Bayes decision rule, Minimum error rate classification, Normal density and discriminant functions (5 Lectures)

Parameter Estimation: Maximum Likelihood and Bayesian Parameter Estimation (2 Lectures)

Fractal II: Unsupervised Learning

Discriminative Methods: Distance-based methods, Linear Discriminant Functions, Decision Tree, Random Decision Forest and Boosting (6 Lectures)

Feature Selection and Dimensionality Reduction: PCA, LDA, ICA, SFFS, SBFS (4 Lectures)

Clustering: k-means clustering, Gaussian Mixture Modeling, EM-algorithm (4 Lectures)

Fractal III: Kernels and Neural Networks

Kernel Machines: Kernel Tricks, SVMs (primal and dual forms), K-SVR, K-PCA (6 Lectures)

Artificial Neural Networks: MLP, Backprop, and RBF-Net (4 Lectures)

Foundations of Deep Learning: DNN, CNN, Autoencoders (4 lectures)

Text Book

1. Shalev-Shwartz,S., Ben-David,S., (2014), Understanding Machine Learning: From Theory to Algorithms, Cambridge University Press
2. R. O. Duda, P. E. Hart, D. G. Stork (2000), Pattern Classification, Wiley-Blackwell, 2nd Edition.

Reference Book

1. Mitchell Tom (1997). Machine Learning, Tata McGraw-Hill
2. C. M. BISHOP (2006), Pattern Recognition and Machine Learning, Springer-Verlag New York, 1st Edition.

Self-Learning Material

1. Department of Computer Science, Stanford University, https://see.stanford.edu/Course/CS229

Course Title	**Mobile and Pervasive Computing**	Course No.	**CSL7460**
Department	Computer Science and Engineering	Structure (L-T-P [C])	3-0-0 [3]
Offered for	B. Tech and M. Tech	Type	Elective
Pre-requisites	Computer Networks	Antirequisite	None

Objectives

1. Explain necessary concepts to understand challenges in developing applications for mobile devices, when contrasted with that of a general-purpose computer.
2. Introduce mechanisms employed by different cellular technologies, particularly 5G, and contrast them with each other.

Learning Outcomes

At the end of this course, students would be able to:

1. Describe the working principles of various cellular technologies, as well as appreciate the associated challenges.
2. Compare/contrast various aspects of Mobile and Pervasive Computing, viz. localization, sensing, security and privacy concerns, application development, etc.
3. Implement the theoretical concepts into practical scenarios..

Contents

Overview of Wireless Systems: Infrastructure-based vs Ad-hoc, Wireless LANs, Cellular systems, Sensor networks, Bluetooth, WiFi, WiMAX. (3 Lectures)
Medium Access: Link Adaptation, Routing Protocols. (4 Lectures)
Mobility and Handoff Management*:* Link layer mobility mechanisms (location management protocols), Network layer mobility mechanisms (Macro and Micro mobility protocols), Handoff management protocols. (6 Lectures)
Cellular Networks: LTE and 5G overview, 5G Architecture, RAN and dynamic CRAN, Mobility management and Network slicing in 5G. (11 Lectures)
Pervasive Computing: Principles, Characteristics, Pervasive devices, Smart sensors and actuators, Context communication and access services. (4 lectures)
Context Aware Sensor Networks: Open protocols (SDP, Jini, SLP, UPnP), SyncML framework, Context aware mobile services, Context aware security. (4 lectures)
Energy Efficiency, Localization **(GPS/WiFi/GSM),** ***Security*** (4 Lectures)
Recent Advances in Wearable devices (3 Lectures)
Body Area Networks **(BAN)** (3 Lectures)

Textbooks

1. I. STOJMENOVIC (2002), Handbook of Wireless Networks and Mobile and Pervasive Computing, Wiley.
2. S. LOKE (2006), Context-aware Pervasive Systems: Architectures for a New Breed of Applications, CRC Press.
3. A. OSSEIRAN, J.F. MONSERRAT, P. MARSCH, (Eds.), 5G Mobile and Wireless Communications Technology, Cambridge University Press, 2016.

Reference Books

1. R. KAMAL (2008), Mobile and Pervasive Computing, Oxford University Press, 3rd Edition.
2. L. MERK, M. NICLOUS (2006), Principles of Mobile and Pervasive Computing, Dreamtech Press, 2nd Edition.
3. G. AGGELOU (2004), Mobile Ad hoc Networks: From Wireless LANs to 4G Networks, McGraw-Hill Professional.

Self-learning Material:

1. J.J. DRAKE, Z. LANIER, C. MULLINER, P.O. FORA, S.A. RIDLEY, G. WICHERSKI (2014), Android Hacker's Handbook, John Wiley, 1st Edition.
2. P. SINGH, NPTEL, IIIT Delhi: https://nptel.ac.in/courses/106106147/
3. Relevant research papers from MobiSys, Sensys, MC2R, MobiCom and UbiComp.

Title	**Natural Language Understanding**	Number	CSL7640
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech, M.Tech., PhD	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. To provide insights into fundamental concepts and algorithms related to Natural Language Understanding
2. Impart working expertise by introducing practical problems.

Learning Outcomes

The students are expected to have the ability to:

1. Formulate natural language understanding tasks
2. Design and implement basic applications of NLU

Contents

Traditional NLU: Introduction to NLU, Motivation, Morphology, Parts-of-Speech, Language Models, Word Sense Disambiguation, Anaphora Resolution, Basics of Supervised and Semi-supervised Learning for NLU, Hidden Markov Models for language modeling, EM Algorithm, Structured Prediction, Dependency Parsing, Topic Models, Semantic Parsing, Sentiment analysis. (14 Lectures)

Deep Learning for NLU: Intro to Neural NLU, Word Vector representations, Neural Networks and backpropagation -- for named entity recognition, Practical tips: gradient checks, overfitting, regularization, activation functions, Recurrent neural networks -- for language modeling and other tasks, GRUs and LSTMs -- for machine translation, Recursive neural networks -- for parsing, Convolutional neural networks -- for sentence classification, Question answering and dialogue system, Graph Neural Network for NLU, Natural Language Generation, Analysis and Interpretability of Neural NLU. (22 Lectures) Know ledge Graphs: Knowledge graph embedding techniques, Inference on knowledge graphs. (6 Lectures)

Textbook

1. C. MANNING, H. SCHÜTZE (1999), Foundations of Statistical Natural Language Processing, MIT Press.
2. D. JURAFSKY, J.H. MARTIN, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition (3rd Edition Draft), 2019.

Reference Books:

1. E. BENDER (2013), Linguistic Fundamentals for NLP, Morgan Claypool Publishers..
2. J. ALLEN (1995), Natural Language Understanding, Pearson Education, 1995.
3. Research Literature.

Self-Learning Material

1. http://web.stanford.edu/class/cs224n/index.html#schedule (Deep learning for NLP)